

LESSON 3-6 • SESSION 2 • PRACTICE • FORM A**Metric Unit Conversions**

Learning target – I can convert between metric units by multiplying or dividing by powers of ten. · 6.RP.A.3d

REMEMBER

- Metric units use **powers of 10**: $1 \text{ km} = 1,000 \text{ m}$ · $1 \text{ m} = 100 \text{ cm}$ · $1 \text{ cm} = 10 \text{ mm}$ · $1 \text{ kg} = 1,000 \text{ g}$ · $1 \text{ L} = 1,000 \text{ mL}$.
- Big unit to small unit: **multiply** ($3 \text{ kg} = 3,000 \text{ g}$).
Small to big: **divide** ($4,000 \text{ g} = 4 \text{ kg}$).
- Make the units **match** before you divide or compare.

WORKED EXAMPLE

A bag has 2.4 kg of flour. How many 300 g portions can you make?

Step 1 Match units: $2.4 \times 1,000 = 2,400 \text{ g}$.**Step 2** Divide: $2,400 \div 300 = 8$.**Step 3** Check: $8 \times 300 = 2,400 \text{ g}$.**Answer 8 portions****WORDS** **metric units** (*unidades métricas*) · **powers of 10** (*potencias de 10*) · **convert** (*convertir*)**PRACTICE***show your work · label units***1.** Convert each measure.

a. $7 \text{ km} = \underline{\hspace{1cm}} \text{ m}$

b. $1,500 \text{ mL} = \underline{\hspace{1cm}} \text{ L}$

c. $3 \text{ kg} = \underline{\hspace{1cm}} \text{ g}$

d. $250 \text{ cm} = \underline{\hspace{1cm}} \text{ m}$

e. $2,450 \text{ m} = \underline{\hspace{1cm}} \text{ km}$

f. $2.5 \text{ mL} = \underline{\hspace{1cm}} \text{ L}$

2. Water Cooler. A cooler has 4 L of water. Complete the table. How many 500 mL bottles can it fill?**Hint** Divide the total milliliters by 500.

Liters	1	2	3	4
Milliliters	1,000	2,000	3,000	

Answer _____ bottles**3. 5K Run.** A charity race is 5 km long. ($1 \text{ km} = 1,000 \text{ m}$)

a. Meters: _____ m

b. A runner has gone 3,200 m. Meters left: _____ m

*Show your work.***4. Lab Mass.** A student measures 350 g of salt. ($1 \text{ g} = 1,000 \text{ mg}$, $1 \text{ kg} = 1,000 \text{ g}$)

a. Milligrams: _____ mg

b. Kilograms: _____ kg

*Show your work.***5. Lab Ribbon.** A teacher cuts a 2.4-meter ribbon into 30 cm pieces. ($1 \text{ m} = 100 \text{ cm}$)**Hint** Change meters to centimeters before you divide.

a. Ribbon length in centimeters: _____ cm

b. Number of 30 cm pieces: _____ pieces

Show your work.

- 6. Backpack Limit.** A backpack weighs 4,800 grams. The safety limit is 5 kg. Is the backpack under the limit? (1,000 g = 1 kg)

Hint Change grams to kilograms: divide by 1,000.

Show your work.

Yes or no _____

- 7. Recycling Bin.** A full recycling bin has a mass of 12.75 kilograms. What is its mass in grams? (1 kg = 1,000 g)

Hint Kilograms to grams is big to small: multiply.

Show your work.

Answer _____ g

- 8. Class Relay.** Each of 8 runners covers 450 meters in a relay.

- a. Total distance in meters: _____ m
- b. Total distance in kilometers: _____ km
- c. Which unit would you use to report the total? Why? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Daniel's Work.** Daniel converted 3 meters to centimeters by dividing: $3 \div 100 = 0.03$ cm. Explain his mistake and find the correct length.

TEST PRACTICE

choose the best answer

- 10.** 4 kilometers equals how many meters?
A 4,000 **B** 400 **C** 40 **D** 0.004
- 11.** Metric conversions all use powers of –
A 2 **B** 5 **C** 12 **D** 10

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

3.6	12,750	0.35	245	1.5	1,275	1,800	8	35	2.45
L	A	M	Z	D	Q	I	C	X	E

#1b	#1e	#2	#3b	#4b	#7	#8b

CHECK YOURSELF

☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____